Foundation of Blockchain Technology (IS501)
Pre-Reading Material – Lecture 29

Encryption Schemes-I

Estimated Reading Time: 15–20 minutes

Introduction to Encryption

Encryption is a cryptographic process that converts readable information, called plaintext, into an unreadable form, called ciphertext. It helps protect information from unauthorized access.

In blockchain technology, encryption can be used to protect confidential information associated with transactions, communications and stored data. However, most public blockchains do not encrypt all transaction data by default; much of their data is publicly visible.

What Is an Encryption Scheme?

An encryption scheme is a method of converting plaintext into ciphertext using a cryptographic algorithm and a key. The ciphertext can be converted back into plaintext through decryption by an authorized party with the appropriate key.

The main components of an encryption scheme are:

- Plaintext: The original readable information.
- Encryption Algorithm: The procedure used to transform plaintext into ciphertext.
- Key: A value used by the encryption or decryption process.
- Ciphertext: The encrypted, unreadable representation of the information.
- Decryption: The process of recovering plaintext from ciphertext.

Types of Encryption Schemes

A. Symmetric Encryption

Symmetric encryption uses the same secret key for encryption and decryption.

The sender encrypts the information using a secret key, and the authorized receiver uses the same key to decrypt it.

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Example: Advanced Encryption Standard (AES).

Advantages:

- Fast and efficient for large amounts of data.
- Suitable for encrypting files and transaction-related information.

Limitation: The secret key must be shared and stored securely.

B. Asymmetric Encryption

Asymmetric cryptography uses a pair of mathematically related keys: a public key and a private key. In encryption-based systems, information encrypted with the recipient's public key can generally be decrypted using the corresponding private key.

Advantages:

- Supports secure communication without sharing the private key.
- Helps establish secure communication channels.

Limitation: It is generally more computationally expensive than symmetric encryption.

Important distinction: Public-key encryption and digital signatures serve different purposes. Encryption primarily protects confidentiality, while digital signatures help verify authenticity and integrity.

Role of Encryption in Blockchain

Encryption can support blockchain applications in several ways:

- Protecting confidential information exchanged between users.
- Securing communication between applications and services.
- Protecting sensitive records stored outside a public blockchain.
- Supporting privacy-preserving blockchain applications.

Blockchain systems also use hashing and digital signatures, but these are not the same as encryption. Hashing supports data integrity, while digital signatures support transaction authorization and verification.

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Example

Suppose Alice wants to send a confidential document to Bob. Alice encrypts the document using an appropriate encryption scheme. Bob uses the required decryption key to recover the original document.

If the encrypted document is stored on a blockchain, only someone with the appropriate decryption key should be able to read its contents, assuming the encryption and key management are secure.

Conclusion: Encryption schemes help protect the confidentiality of digital information. Symmetric and asymmetric encryption provide different approaches to securing data, and both can support privacy requirements in blockchain applications.

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